Great ships will arrive carrying essential technology from Earth, new families of settlers, scarce resources. It is hard to know, on the basis of our limited knowledge of Mars, whether they will go home empty—or whether they will carry with them something found only on Mars, something considered very valuable on Earth. Initially much of the scientific investigation of samples of the Martian surface will be done on Earth. But in time the scientific study of Mars (and its moons Phobos and Deimos) will be done from Mars.

Eventually—as has happened with virtually every other form of human transportation—interplanetary travel will become accessible to people of ordinary means: to scientists pursuing their own research projects, to settlers fed up with Earth, even to venturesome tourists. And of course there will be explorers.

If the time ever came when it was possible to make the Martian evironment much more Earth-like—so protective garments, oxygen masks, and domed farmlands and cities could be dispensed with—the attraction and accessibility of Mars would be increased many-fold. The same, of course, would be true for any other world which could be engineered so that humans could live there without elaborate contrivances to keep the planetary environment out. We would feel much more comfortable in our adopted home if an intact dome or spacesuit weren't all that stood between us and death. (But perhaps I exaggerate the dangers. People who live in the Netherlands seem at least as well adjusted and carefree as other inhabitants of Northern Europe; yet their dikes are all that stand between them and the sea.

Recognizing the speculative nature of the question and the limitations in our knowledge, is it nevertheless possible to envision terraforming the planets?

We need look no further than our own world to see that humans are now able to alter planetary environments in a profound way. Depletion of the ozone layer, global warming from an increased greenhouse effect, and global cooling from nuclear war are all ways in which present technology can significantly alter the environment of our world—and in each case as an inadvertent consequence of doing something else. If we had *intended* to alter our planetary